



Building the TReMeDa Datasets

A Research Compendium

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Version: Development draft
Produced: 2026-06-11

Site repository: <https://github.com/CGMoreh/tremeda-compendium-site>
Source code: <https://github.com/CGMoreh/tremeda-compendium> (Private repo – request access to contribute)

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Introduction

The TReMeDa project

The Trust Research Methodology Database (TReMeDa) is building a curated database of metadata and associated replication data for English-language publications on the topic of ‘social trust’ in the social sciences, broadly construed, over the three decades since 1990.

“Social trust” is treated here as a broad concept that spans many definitions and operationalisations of trust in a social-science context – not only generalised trust, but interpersonal and institutional trust as well. That breadth is deliberate, and it is also the central methodological difficulty: a publication may carry an important trust component without naming it in its title or abstract, and the bare word “trust” is often used as shorthand whose precise sense is supplied by the journal or discipline. The most encompassing discovery strategy is therefore to capture everything that mentions trust in title, abstract or keywords, and then use further evidence – textual, publication-level and disciplinary – to decide whether a given output really concerns the kind of trust the project is about.

The three datasets

The database is organised as three nested subsets, each serving different aims.

THEME – the thematic dataset. The master layer: general bibliometric metadata for trust research in the social sciences, drawn from Web of Science, Scopus, and OpenAlex, and augmented with classifications that can be extracted at acceptable reliability by machine-learning tools (for example a broad methodological split into theoretical, empirical-qualitative, and empirical-quantitative work). Expected raw size of ~75,000+ entries, plus manual additions and backfills, minus exclusions due to mis-identification on technical grounds (i.e. not quality-driven).

META – the meta-analytical dataset. A restriction of THEME to primary quantitative empirical studies, augmented with deeper descriptors: primary versus secondary data sources, whether trust is the dependent or independent variable, reproducibility and data-access statements, statistical methods, and causal claims. Expected scale: between a few hundred and ~2,000 publications maximum.

METHOD – the methodological dataset. The smallest and most detailed layer: studies whose findings can be replicated from publicly available data or author-provided replication packages, accompanied by documented analytical datasets and annotated code. Its purpose is pedagogical – to underpin a textbook on quantitative sociological methodology applied to researching trust, and to serve instructors more generally. Expected scale: around 50-100 publications spanning a broad range of disciplines and methods, broken down into a larger number of analytical datasets designed to demonstrate individual methods and/or published outputs.

What these notebooks document

These notebooks are the running, cleaned record of those methodological audits and the decisions they support. They are organised by the three datasets above. Work to date has concentrated on **THEME**; the **META** and **METHOD** parts are placeholders that will be filled as the project moves downstream.



THEME – the thematic dataset

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Overview

The thematic dataset is the foundation the other two are carved from, so the first methodological task is to settle how it is discovered – which sources to search, and how. The project does not begin by assuming that any one database is sufficient; it begins from what is already known.

A background notebook (Notebook 1.) reviews what the literature and OpenAlex’s own documentation show about the bibliographic databases available for this work: that abstract metadata is being withdrawn from open databases at scale, and that OpenAlex’s subject labels are an automated, AI-built artefact rather than an intrinsic property of a paper. That context sets the terms for a practical comparison.

Which sources, and can one suffice?

The conventional approach combines several subscription databases – expensive to maintain and awkward to keep current. OpenAlex is an attractive single source – open, free, very large, and queryable by API – but only if it does not *miss* what the curated databases (Scopus, Web of Science) would catch. The way to find out is to compare them directly. A focused single-year (2015) audit did this across the three most authoritative bibliographic databases the team has institutional access to, verifying every apparent gap against the live databases to separate a genuine coverage gap from a mere retrieval gap; it is summarised in section 3.1.

The audit reaches a layered answer. The OpenAlex *database contains* almost everything the curated searches return, so coverage is not the obstacle; but no single query *retrieves* it all, because each database draws its disciplinary and document-type boundaries differently. Discovery therefore has to draw on all three sources together, merged by identifier and filtered downstream.

The databases notebook also sets out the **selection design** built on that answer – the curated Scopus and Web of Science keyword searches, the OpenAlex twelve-topic union, and the by-DOI backfill that recovers curated metadata for the only-OpenAlex works (section 1.4, section 1.5). A further notebook describes the record-level **trust classifier** that triages each retrieved record (Notebook 2.); the assembled **2000–2025 corpus** is reported and interrogated in Notebook 3.; the construction **pipeline** is summarised visually in Notebook 4.; and every column of the dataset is catalogued in the **codebook** (Notebook 5.). A disciplinary-core test and a citation reference miner for sources that keyword search cannot see are planned for a later stage, building on the same evidence.

1. Bibliographic databases: Web of Science, Scopus, OpenAlex

i Status: background + the implemented selection design

This notebook does two things. First it sets out what is already known – from the literature and from each provider’s own documentation – about the three bibliographic databases this project relies on: their coverage, their search tools, and the keyword and classification systems they add. Then it sets out the **selection design** the project actually ran on them, and the curated by-DOI backfill that completes it. The empirical corpus that results is the subject of Notebook 3..

The thematic dataset is discovered from three bibliographic databases – **Web of Science**, **Scopus** and **OpenAlex** – chosen because they are the three most authoritative sources the team has institutional access to, and because they are complementary rather than interchangeable. This notebook first describes what each one is and what it adds beyond a list of references (its search tools and its keyword/classification systems), then turns to the issue that shapes every later design choice – the metadata we query, the abstract above all, is not stable – and finally sets out how the project searches all three and reconciles the result.

1.1 What the three databases are, and what they add

All three index scholarly works and let us search them, but they are built differently, cover the literature differently, and – crucially for a keyword project – attach different *added* metadata to each record. The differences are not incidental; they are the reason no single source suffices.

1.1.1 Web of Science

Web of Science (Clarivate) is the oldest and the most **selective** of the three: its Core Collection is an editorially curated set of journals chosen for citation impact, so it is comparatively clean but narrower, and its coverage skews towards established English-language journals. For discovery, its **TS=** (“topic”) field reaches four fields at once – title, abstract, **author keywords**, and **Keywords Plus**. Keywords Plus is the distinctive addition: index terms generated **algorithmically by Clarivate from the titles of the work’s cited references**, not supplied by the author. Because they are derived from what a paper *cites* rather than what it *says*, Keywords Plus can surface a study on a term it never uses itself – a quantitative trust paper that cites the trust canon heavily can be retrieved on “trust” even when its own title and abstract never mention it. Web of Science also assigns each work the subject **Categories** and **Research Areas** of its journal – a journal-level classification, not a per-article one.

1.1.2 Scopus

Scopus (Elsevier) is **broad**: it indexes more journals, conference proceedings and book series than Web of Science, so it returns more, at some cost in selectivity. Its **TITLE-ABS-KEY** field reaches title, abstract, **author keywords** and **index keywords**. The index keywords are Scopus’s distinctive addition: **controlled-vocabulary terms assigned from standardised thesauri** (for example Emtree for the biomedical literature) rather than chosen by the author – a curator/algorithm-assigned layer that normalises terminology across a field. Scopus classifies each source under the **All Science Journal Classification** (ASJC: 27 subject areas, ~300 minor categories) – again a journal-level scheme. The practical effect is that Scopus is the broadest keyword net of the three and the one most likely to return the cross-disciplinary tail.

1.1.3 OpenAlex

OpenAlex (Priem Et al., 2022) is the newcomer and the structural departure: a **fully open**, free, very large index (over 250 million works) launched in 2022 by the OurResearch non-profit as a successor to the discontinued Microsoft Academic Graph, built on open data, an open API and open-source code. Two features make it attractive as a backbone. It is open and queryable at scale without a subscription, so it can serve as the discovery substrate and citation graph for a project of this size. And rather than rely only on journal-level subject categories, it assigns **each work its own machine-assigned topic** from a citation-built hierarchy – the genuinely novel, LLM-infused addition that the next section unpacks, because three of this project’s design choices turn on how much weight those topic labels can bear. The trade-off is that OpenAlex is the least selective and, as the rest of this notebook shows, the most exposed to the metadata instability that now threatens keyword discovery.

1.2 How OpenAlex builds its topics, and what it offers

Because the OpenAlex topic system is both the reason to use it and a thing to be cautious about, it is worth setting out exactly how those labels are made. The current scheme, **Topics**, is documented in the CWTS team’s open whitepaper and dataset (Eck, 2024; Eck & Waltman, 2024) and in OpenAlex’s own end-to-end process notes: a four-step pipeline whose every label is the output of stacked automated layers, none of them systematically audited.

1. **Cluster.** CWTS clustered the ~71 million OpenAlex works (2000–2023) that carry incoming or outgoing citations – linked by ~1.7 billion citation edges – using the extended direct-citation approach and the Leiden algorithm (Waltman & Eck, 2012; Traag Et al., 2019). This produced **4,521 micro-level** research areas, grouped into **917 meso** and **20 macro** areas. The clusters are defined purely by citation structure: no text, no keywords.
2. **Label.** To name 4,521 clusters without manual effort, CWTS fed the **titles of a sample of each cluster’s most-cited papers** to a large language model (an updated **GPT-3.5 Turbo**; the open labelling tool also supports GPT-4) which returned, per cluster, a short label, a long label, ten keywords, a summary and a Wikipedia URL (Eck, 2024). The authors note that they did not systematically evaluate label quality and that the same approach failed at higher aggregation levels. The labels, keywords and summaries in the mapping table are this model’s output.
3. **Map to a hierarchy.** OpenAlex then placed each citation-defined topic into the Scopus **ASJC** structure (4 domains, 26 fields, 254 subfields) that it exposes as `domain / field / subfield`, using a *second* LLM pass, because asking a person to place over 4,000 topics by hand was impractical. So the `domain` we filter on is not an intrinsic property of a paper – it is an LLM’s judgement about which ASJC bin a citation cluster’s machine-written label most resembles, and the team flags that some topics could sit in several subfields.
4. **Classify works.** Finally, to tag the ~250 million works – only a third of which carry citation data and only about half a usable abstract – OpenAlex trained a supervised model on the CWTS labels: a fine-tuned multilingual BERT over the merged title-and-abstract text, fed into a network that also takes a journal-name embedding and “gold citation” features. The model was trained with features randomly held out so it degrades gracefully when the abstract or the citations are missing. The per-work classifier is therefore *not* an LLM: the only LLM steps write the bin names (step 2) and file the bins into the hierarchy (step 3).

Two properties matter for us. First, the per-work label is **single** (`primary_topic`) and is assigned from four signals of uneven reliability – title (~90% of works), abstract (~50%), journal name, and citations (~33%). Second, because the model was built to tolerate a missing abstract, abstract withdrawal does not break topic assignment outright; it removes the richest *text* signal and pushes the model onto title, journal and citation features. That fallback is adequate for placing a paper in broad subject space, but it is precisely the signal that cannot separate the *senses* of trust: a closed-access paper titled “Antecedents of relationship commitment,” stripped of its abstract, is placed by its journal and its citations, not by whether its trust is interpersonal, institutional or brand trust.

1.2.1 Where “trust” lives in the OpenAlex topic map

The mapping table also settles a question we had set aside: is there a single natural “trust topic” to scope on? There is not. Searching all topics’ labels, keywords and summaries for trust returns **27 topics**, spread across three domains (21 Social Sciences, 5 Physical Sciences, 1 Health Sciences) and many fields. Only one

carries “trust” in its actual label – topic 10927, “Access Control and Trust” – and it is an online-reputation and computer-security cluster filed, through the step-3 mis-mapping above, under the Sociology and Political Science subfield: a clean instance of exactly the off-target trust this project wants to exclude. The trust the project *does* want is scattered across a handful of topics in different fields, several of which do not announce trust in their label at all (social capital, experimental behavioural economics, risk perception, e-government, policing, vaccine hesitancy, oxytocin-and-trust, culture-and-development). No single-topic scope can work, and scoping by topic *label* or *subfield name* is unsafe; the defensible middle path is a **curated union of inspected topics**, which is exactly what the selection design (section 1.4) uses.

Set against its construction, the OpenAlex topic system offers a great deal as **input** and very little as **verdict**. As input it is excellent: a free, citation-grounded clustering of the whole literature, with per-work labels robust to missing metadata. As a verdict it is weak for our purpose: single-label, three automated layers deep with no systematic audit, its domain placement an LLM’s guess about ASJC bins, and its discriminating text signal the very abstract now being withdrawn. The reasonable use is to lean on it for recall and structure and to reserve the in/out and trust-sense judgements for our own content-level screen (Notebook 2.). Comparative work on OpenAlex coverage and metadata completeness reaches the same posture – strong as an open substrate, uneven enough to verify rather than trust blindly (Alonso-Álvarez & Eck, 2025).

1.3 Why we cannot rely on OpenAlex alone

A discovery design that leans on OpenAlex uses the abstract in three places at once: the keyword clause that finds candidate papers, any downstream abstract-level screen, and OpenAlex’s own abstract-fed topic labels. Anything that degrades the abstract degrades all three at once – and the degradation is not uniform: it falls hardest on closed-access, large-publisher, recent work, disproportionately the high-quality quantitative studies the project most wants to keep.

Abstracts are being withdrawn, and the incentive is AI. Aaron Tay’s (2025) analysis frames open abstracts as “the petrol tank for AI discovery” and measures the drain: Elsevier abstract coverage in OpenAlex fell from roughly 82% in November 2024 to 22.5% by September 2025, Springer Nature from about 50% to 35.8%. The OpenAlex maintainers describe the same process from inside – many abstracts were inherited from Microsoft Academic Graph without an open licence and have been removed on copyright grounds (Springer Nature, then ACS, OUP, Wiley, Cell Press, Elsevier and others). Now that AI companies will pay for training text, publishers have re-classified the abstract as a strategic asset and are closing the indirect channels open aggregators relied on, so the withdrawal will not simply reverse. Stansfield et al. (2025), testing OpenAlex for systematic reviews, found previously retrievable studies became unfindable once their abstracts were removed, with recall drifting downward as the corpus changed underneath them.

The open-abstract commons was never full. The Initiative for Open Abstracts [IOA; Initiative for Open Abstracts (2020)] asks publishers to deposit abstracts openly in Crossref; it succeeded with many smaller and fully-open publishers but not with the largest closed ones, and only on the order of 7% of Crossref works carry a deposited abstract. The open channel that should have absorbed the loss was always partial, and the scraped channel that masked that partiality is now closing.

Even the abstracts that remain are partly corrupt. Kim, Holst and Ginis (2026) assessed 10,000 sampled OpenAlex abstracts and found **12% of records flagged `has_abstract=True` do not contain a valid scientific abstract** (PubMed conclusion-only snippets, repository metadata, web-scrape chrome, truncation, placeholders, wrong genre). The rate falls over time but the failures cluster by era, source and document type, so they bias rather than merely add noise.

For a baseline, Culbert et al. (2025) found OpenAlex abstract coverage of about 87% against over 92% for Web of Science and Scopus. The gap to the curated databases was already real before the latest withdrawals; it is widening month by month, and specifically for the closed-access, large-publisher, recent content the project cares about. **That is the case for using all three sources rather than OpenAlex alone:** the curated databases hold abstracts and curated keyword fields that OpenAlex is losing, while OpenAlex holds the open citation graph and the per-work topic net the curated databases lack. Each covers the others’ blind spots.

1.4 The selection design

Because no single query is both precise and complete, the project operationalises “the trust we want” at **three moments**, each with different strengths and blind spots, and combines them:

1. **Query-time selection in the curated databases** – a deliberately precise keyword design run on Scopus and Web of Science.
2. **Query-time selection in OpenAlex** – selection by the work’s machine-assigned *topic* rather than by words, over a curated union of trust-bearing topics.
3. **Record-level classification after retrieval** – a transparent rule that reads each retrieved record and labels the *kind* of trust it is about.

The first two decide what enters the corpus and are set out here; the third – the trust-sense classifier – is the subject of Notebook 2.. All queries are archived verbatim in [data/queries/](#).

1.4.1 Selection in the curated databases: a precise keyword design

Scopus and Web of Science are searched with the same logic in each database’s syntax: a block of explicit **sense phrases** (“social trust”, “political trust”, “institutional trust”, ...), a **proximity block** pairing the trust word family with the objects of social-scientific trust (government, institutions, police, courts, media, science, experts, strangers, people, citizens, ...) within three words, and a guarded **confidence-in-institutions** clause that picks up the survey-research idiom (“confidence in parliament”) without admitting the everyday sense of confidence. “Social capital” is deliberately excluded here, because it re-enters through the OpenAlex topic union and the reference miner; “trustee”/“trusteeship” are kept out by enumerating the trust terms rather than truncating *trust**. There is no discipline restriction – breadth is cross-disciplinary, and precision is imposed later by the classifier and the screen. The range is 1999–2026 (buffer years around the 2000–2025 analysis window), English, articles and book chapters.

Scopus (Advanced Search):

```
TITLE-ABS-KEY(
  ( {social trust} OR {political trust} OR {institutional trust} OR {interpersonal trust}
    OR {generalized trust} OR {generalised trust} OR {particularized trust} OR
    {particularised trust}
    OR {dispositional trust} OR {system trust} OR {civic trust} OR {personal trust}
    OR {general trust} OR {public trust}
    OR {political distrust} OR {institutional distrust} OR {social distrust} OR
    {public mistrust} )
  OR ( (trust OR trusted OR trusting OR trusts OR trustworth* OR mistrust* OR distrust*)
    W/3 (government* OR governance OR parliament* OR politician* OR institution* OR
    police OR court*
      OR judiciary OR authorit* OR media OR science OR scientist* OR expert*
    OR stranger*
      OR others OR neighbour* OR neighbor* OR people OR citizen* OR society
    OR state) )
  OR ( confidence W/3 (government* OR parliament* OR institution* OR police OR court*
    OR politician* OR authorit*) )
)
AND PUBYEAR > 1998 AND PUBYEAR < 2027 AND LANGUAGE(English) AND (DOCTYPE(ar) OR
DOCTYPE(ch))
```

Web of Science (Core Collection; TS= reaches title, abstract, author keywords and the cited-reference-derived Keywords Plus; ? is a single-character wildcard):

```
TS=(
  ( "social trust" OR "political trust" OR "institutional trust" OR "interpersonal trust"
    OR "generalized trust" OR "particularized trust" OR "dispositional trust" OR
    "system trust"
    OR "civic trust" OR "personal trust" OR "general trust" OR "public trust"
    OR "political distrust" OR "institutional distrust" OR "social distrust" OR
    "public mistrust" )
  OR ( (trust OR trusted OR trusting OR trusts OR trustworth* OR mistrust* OR distrust*)
    NEAR/3 (government* OR governance OR parliament* OR politician* OR institution*
    OR police OR court*
      OR judiciary OR authorit* OR media OR science OR scientist* OR expert*
    OR stranger*
      OR others OR neighbour* OR neighbor* OR people OR citizen* OR society
    OR state) )
)
```

```
OR ( confidence NEAR/3 (government* OR parliament* OR institution* OR police OR
court* OR politician* OR authorit*) )
)
AND PY=(1999-2026) AND LA=(English) AND DT=(Article OR "Book Chapter")
```

1.4.2 Selection in OpenAlex: by topic and broad keyword search across titles and abstracts

OpenAlex is queried differently, because its strength is the machine-assigned **topic** of each work (section 1.2). Rather than restrict to one topic – which would capture only the trust-*defining* core – the pull uses a curated **union of twelve topics** where trust research concentrates across domains, intersected with the trust keyword family so that off-target members of those topics are filtered at the source. The twelve topics:

Table 1.1: The twelve OpenAlex topics in the discovery union.

Topic ID	What it covers
T10646	Behavioural-economics trust games
T10827	Patient-provider communication
T10833	Vaccine hesitancy
T10953	E-government
T11076	Policing and legitimacy
T11239	Social capital
T11541	Oxytocin / neuro-trust
T11573	Risk perception
T11696	Conflict and negotiation
T12028	Knowledge management
T12121	Cooperative studies
T12312	Culture, economy and development

The live API call (1999–2026, English, articles and book chapters):

```
GET https://api.openalex.org/works?filter=
  primary_topic.id:T10646|T10827|T10833|T10953|T11076|T11239|T11541|T11573|T11696|
  T12028|T12121|T12312,
  title_and_abstract.search:trust OR distrust OR mistrust OR trustworthy OR
  trustworthiness OR distrustful OR mistrustful OR untrustworthy,
  language:en,
  type:article|book-chapter,
  publication_year:1999-2026
```

The keyword search runs on *titles* and *abstracts* (and it seems to pick up keywords present in an abstract even when the abstract is now protected and not included in the pulled metadata). A by-DOI **crosswalk** then fetches the full OpenAlex record for every curated (Scopus/WoS) work not already in the topic pull, so the citation graph and OpenAlex metadata attach to every work. The collection scripts (*Scripts/collect/openalex-topic.R*, *Scripts/collect/openalex-crosswalk.R*) hold the exact parameters.

1.5 Completing the only-OpenAlex region: the curated DOI-backfill

Selection by these three channels leaves one large, awkward group: works the OpenAlex topic search returned but the curated keyword searches did not – the **only-OpenAlex** (pull-only) region. Two terms keep the next step straight. The **OpenAlex pull** is what the topic search returned; the **OpenAlex database** is what OpenAlex holds regardless of any query; symmetrically, the **curated searches** are what the Scopus/WoS keyword queries returned, and the **curated databases** are those indexes in full. A pull-only work was not *returned* by a curated search, but its DOI may nonetheless *exist* in a curated database – the search simply did not match it.

A **reverse check** establishes which: for every pull-only work with a DOI, the live Scopus and Web of Science APIs are asked whether that DOI exists in their database at all ([Scripts/collect/reverse-check.py](#)). The works whose DOI does exist there – present in a curated database but missed by the curated search – are **recoverable**: their abstract and curated keyword fields are sitting in a record the keyword search never returned. A bulk by-DOI pull then fetches those full curated records ([Scripts/collect/backfill-build-queries.py](#) builds the queries; the exports are re-merged in assembly), enriching the pull-only works with curated abstracts, author/index keywords and Keywords Plus, and document types they otherwise lacked.

Two safeguards matter. The backfill is **metadata only**: it fills the `Scopus_*` / `WoS_*` fields and is flagged (`scopus_backfilled` / `wos_backfilled` / `curated_backfill`), but it never sets the retrieval-sense membership flags `in_Scopus` / `in_WoS` – those continue to mean “the *search* returned it”, so every coverage, overlap and added-value result in Notebook 3. is unaffected by the backfill. And it respects the licence boundary: recovered abstracts are tagged licensed and used for local classification, never redistributed and never sent to an external service. What this backfill recovers, and how much of the abstract gap it closes, is reported in Notebook 3..

2. Defining and detecting ‘trust’

i Status: conceptual (scripted and reproducible)

The record-level classifier is `Scripts/trust_context.py`; it is applied in the derive stage (`Scripts/derive/variables.py`) and documented column-by-column in the derivation registry (`Scripts/derivations.yaml`). This notebook describes the classifier on its own terms; what it finds across the assembled corpus is reported in Notebook 3..

2.1 Selection decides what enters; the classifier decides what each record is about

Building a dataset of social-trust research runs into a difficulty the databases notebook (Notebook 1.) sets out and our own experiments kept confirming: **no single way of selecting records is both precise and complete**. A broad subject filter is high-recall but full of off-target trust (consumer trust in brands, user trust in devices, trust in network protocols); a single well-chosen topic is clean but captures only a fraction of the trust literature, because trust is studied *inside* hundreds of substantive topics rather than in one; and a keyword search is precise but blind to work that never says “trust” in the fields the search can reach.

The project’s answer, set out in the databases notebook, is to stop looking for that one query and instead operationalise “the trust we want” at **three moments**: a precise keyword design run on Scopus and Web of Science, a topic-based pull from OpenAlex, and a record-level classification *after* retrieval. The first two – the selection design that decides what enters the corpus – are described there (section 1.4). This notebook is the **third moment**: the transparent rule that reads each retrieved record and labels the *kind* of trust it is about. It is the scaffold every later stage of screening builds on, so it is set out here conceptually; the distribution it produces over the assembled corpus is reported in the corpus chapter (Notebook 3.).

2.2 The trust-sense classifier

The classifier decides what each record *is about*. It is a transparent, rule-based tool (`Scripts/trust_context.py`), not a model: for each record it scans the title, abstract and the merged keyword fields for the trust word family, extracts the **qualifier before** a trust word (“social trust”) and the **object after** trust/confidence/faith/reliance + in/on/of (“trust in government”), and maps each to a fine **trust_sense** drawn from a lexicon grounded in the survey-research literature (the WVS/ESS confidence-in-institutions battery and the standard distinctions: generalised/social, interpersonal, institutional/systems, political, legal/justice, science/experts, media/information, religion, civic/community, and the health-domain clinical/relational sense; against the off-construct senses: consumer/marketing, technology, security/computing, commercial-finance).

A coarse **trust_scope** is then rolled up from that sense, and it is the workhorse for all downstream screening:

- **in-scope** – the dominant matched sense is one of the social-science trust senses (health-domain trust is in-scope).
- **out-of-scope** – the dominant sense is out of the construct (consumer, technology, security, finance).
- **undetermined** – no qualifier or object matched, so the rule did not pin a sense.

Three properties of this scheme matter for how it is used:

1. **The three states are asymmetric.** **in-scope** and **out-of-scope** mean *the rule reached a decision*; **undetermined** means *the rule was silent* – it does **not** mean “no trust” or “irrelevant”. It is the large residual precisely because trust is often discussed without a lexical cue the rules catch, and because some records have only a title to read.

2. **out-of-scope is a lower bound.** An off-construct study whose object is not matched falls into `undetermined`, not `out-of-scope`, so the out-of-scope count understates the true off-construct share. `in-scope`, by contrast, is high-precision: the lexicon is specific.
3. **It is a prior, not a verdict.** The rule is cheap and auditable; it is meant to be a feature and a triage that a human reviewer and, later, a local LLM refine – the `undetermined` set is the screen’s main queue, and any rule decision can be overridden.

2.2.1 Reading the keyword fields, and recording where the sense came from

Because a record can be silent on trust in its title and abstract yet clearly trust-tagged in its index terms – Web of Science’s Keywords Plus, derived from cited references, is a recurring example (Notebook 1.) – the classifier reads the **full merged keyword set** (author keywords, Scopus index keywords, WoS Keywords Plus, OpenAlex keywords), not just title and abstract. To keep that transparent, it classifies the title-and-abstract text and the keyword text **separately** and records the provenance: `trust_sense_text`, `trust_sense_keywords`, a `sense_basis` flag (text / keywords / both / none), and `scope_conflict`, which is set only when the two signals disagree on scope itself (one in-scope, the other out-of-scope). On a genuine scope conflict the **text wins** the assignment – it is the authors’ own framing – and the flag marks the case for review; small sense-only differences within the same scope are not flagged, so they raise no spurious conflicts. How much the keyword fields add, and how the triage distributes across the assembled corpus, is reported in Notebook 3..

2.2.2 A tested exclusion: OpenAlex concepts

One candidate input was considered and rejected on evidence rather than assertion. OpenAlex assigns each work a list of machine-generated **concepts**, which are broad and noisy. Feeding them to the classifier would resolve only about a tenth of a percent of the undetermined works, and it would do so *harmfully*: the works it “resolves” include computing and security studies pulled wrongly in-scope through concepts like “Trust management (information system)” – for instance a blockchain trust-management scheme and an IoT trust-based routing scheme, both of which are exactly the off-construct technical trust the triage should exclude. The negligible gain and the false-positive cost are why the classifier uses the curated keyword fields but not `concepts`. (The experiment is `Scripts/experiments/concepts.py`.)

2.3 The full classifier criteria

So the rules can be examined and reasoned about without reading the code, the complete criteria set is reproduced here (generated from `Scripts/trust_context.py`; the lexicon currently holds 15 senses – 11 in-scope, 4 off-target – and 49 multiword objects).

How a record is read. Two regular expressions scan the lower-cased title + abstract + merged keywords. The first captures the **qualifier immediately before** a trust word (“social trust” yields `social`); the second captures the **object after** a trust word (“trust/confidence/faith/reliance in/on/of X”, up to three words – “trust in the government” yields `government`). Each captured qualifier or object is mapped to a fine **sense** – multiword objects are tested first, then single tokens against the lexicon below, with stopwords skipped and object-only terms counted only as objects. The most frequent matched sense becomes `trust_sense`; `trust_scope` is **in-scope** if that sense is one of the eleven social-science senses, **out-of-scope** if one of the four off-target senses, and **undetermined** if nothing matched.

2.3.1 In-scope senses (`trust_scope = in-scope`)

- **generalised/social** – generalized, generalised, social, diffuse, general, thin, particularized, particularised, horizontal
- **interpersonal** – interpersonal, mutual, dyadic, relational
- **in-people/others** – people, others, strangers, stranger, humanity, neighbours, neighbors, humans, everyone
- **institutional/systems** – institutional, institutions, institution, organizational, organisational, bureaucratic, bureaucracy, system, systems, authorities, authority, regulators, regulatory, agencies, agency, unions, union, ngos, ngo, companies, corporations, corporate, business, banks, bank, abstract, expert, nhs

- **political** – political, government, governmental, politicians, politician, democracy, democratic, state, parliament, parliamentary, regime, electoral, public, council, congress, senate, citizen, parties, party, politics, governance
- **legal/justice** – police, judicial, judiciary, court, courts, legal, justice, law, lawyers
- **science/experts** – science, scientists, scientific, experts, expertise, academia, academic, universities, university, research, researchers, knowledge
- **media/information** – media, news, press, journalists, journalism, information, misinformation, television, newspapers
- **religion** – church, churches, religion, religious, clergy, god, faith, mosque, temple, priests, spiritual
- **civic/community** – civic, community, communal, neighbour, neighbor, neighbourhood, neighborhood, associations, voluntary
- **clinical/relational-health** – patient, patients, physician, physicians, doctor, doctors, nurse, nurses, clinician, clinical, provider, providers, dentist, therapist, medical, health

2.3.2 Off-target senses (**trust_scope = out-of-scope**)

- **consumer/marketing** – consumer, consumers, customer, customers, brand, seller, sellers, vendor, vendors, purchase, shopping, retail, ecommerce, advertising, marketing, buyer, loyalty, marketplace, marketplaces, influencer, influencers
- **technology** – technology, technological, automation, robot, robots, algorithm, algorithms, ai, artificial, autonomous, website, websites, app, apps, chatbot, platform, automated, machine, machines, internet, device, devices, smartphone, gpt, chatgpt, llm, llms, generative, metaverse, driverless, drone, drones, wearable, wearables, recommender
- **security/computing** – computing, access, cyber, blockchain, cryptographic, protocol, node, nodes, iot, cloud, sensor, sensors, software, wireless, authentication, encryption, phishing, malware, biometric, biometrics
- **commercial-finance** – financial, investor, investors, fintech, credit, insurance, crypto, cryptocurrency, blockchain, market, trading, bitcoin, stablecoin, defi, derivatives

2.3.3 Multiword objects (matched as substrings of the captured object, before single tokens)

- **institutional/systems** – health system, healthcare system, health care system, public health, health authorities, education system, welfare state, welfare system, civil service, armed forces, public institutions, public sector, public administration, european union, united nations, world bank, international organizations, banking system, financial system, financial institutions
- **legal/justice** – criminal justice, justice system, legal system, rule of law
- **political** – political parties, political system, local government, central government, national government
- **science/experts** – scientific community, higher education
- **media/information** – social media, mass media, news media
- **technology** – machine learning, deep learning, neural network, large language model, language model, generative ai, self driving, autonomous vehicle, autonomous vehicles, recommender system, recommendation system, virtual assistant
- **security/computing** – smart contract
- **consumer/marketing** – social commerce, e commerce

The off-target multiwords are deliberately multiword-only where a single token would collide with an in-scope sense: “deep learning” and “neural network” are off-target, but bare **deep** (“deep trust”) and **neural** are not in the lexicon, and **recommendation** (“expert recommendation”) is reached only through “recommender system” / “recommendation system”.

2.3.4 Special handling

- **object-only** (counted only as the object of “trust in X”, never as a qualifier, so “an NHS Trust” – an organisation – is not read as institutional trust): **nhs**.
- **stopwords** (skipped as qualifiers and objects): a, an, and, as, based, between, both, build, building, built, each, for, her, high, his, how, in, is, its, lack, less, level, loss, low, more, of, on, other, our, s, such, than, that, the, their, this, to, very, what, with.

💡 The core algorithm (Scripts/trust_context.py)

```
# qualifier directly before a trust word ("social trust" -> "social")
MODIFIER = re.compile(r"\b([a-z]+\s+(?:dis|mis|un)?trust(?:worthy|worthiness|ed|ing)s)?\b", re.I)
# object after trust/confidence/faith/reliance/belief in|on|towards|of (up to 3 words)
OBJ = re.compile(r"\b(?:dis|mis|un)?trust(?:ed|ing)s?|confidence|faith|reliance|belief)"
r"\s+(?:in|on|towards?|of)\s+(?:the\s+|their\s+|our\s+|a\s+|an\s+|its\s+)?"
r"([a-z]+(?:\s+[a-z]+){0,2})", re.I)

def sense_of(phrase):
    for mw, s in MULTIWORD.items(): # multiword objects win
        if mw in phrase:
            return s
    for tok in phrase.split(): # else first single token in the lexicon
        if tok in KW2SENSE:
            return KW2SENSE[tok]
    return None

def classify(text):
    t = str(text).lower()
    mods = [m.group(1).strip() for m in MODIFIER.finditer(t)]
    objs = [m.group(1).strip() for m in OBJ.finditer(t)]
    hits, kept = Counter(), []
    for p in mods: # qualifier before trust ("social trust")
        if p in STOP or p in OBJECT_ONLY:
            continue
        s = sense_of(p)
        if s: hits[s] += 1; kept.append(p)
    for p in objs: # object after "trust/confidence in X"
        if p in STOP:
            continue
        s = sense_of(p)
        if s: hits[s] += 1; kept.append(p)
    return (hits.most_common(1)[0][0] if hits else "unspecified"), kept
```

2.4 Limitations and what builds on this

The classifier is deliberately simple, and its limits are the reasons it is a *first* layer rather than the last word: it depends on the abstract (a title-only record has little to read), it takes the most frequent matched sense rather than weighing them, its lexicon is finite, and *out-of-scope* is a lower bound. None of this is a defect for its purpose – a transparent, reproducible triage and feature – but it is why the design treats the rule’s output as a prior. The *undetermined* set and any contested decisions are the queue for the human reviewers and for a local large-language-model classifier developed against this same vocabulary; that model is built and validated on openly-licensed abstracts and run in production over the licensed material by a local open-weight model, so it never sends licensed text to an external service. The rule labels everything cheaply and auditably; the later stages spend their effort where the rule is silent or contested.

2.5 Reproducibility

The classifier is `Scripts/trust_context.py`, applied by `Scripts/derive/variables.py` and documented column-by-column in `Scripts/derivations.yaml` (rendered in the codebook, Notebook 5.). The selection queries that decide what enters the corpus are set out in the databases notebook (section 1.4) and

archived verbatim in [data/queries/](#). The concepts experiment is [Scripts/experiments/concepts.py](#). The distribution the classifier produces over the corpus is reported and tabulated in Notebook 3..

3. The THEME corpus

i Status: scripted and reproducible

The corpus is built by the pipeline in `Scripts/` (collection → `assemble/1-spine.py` → `assemble/2-abstracts.py` → `derive/variables.py`), which writes the analysis dataset `data/final/corpus_derived.parquet`. Every table and figure in this notebook is computed from that `corpus_derived` dataset, imported via the R `arrow` package.

3.1 What the 2015 audit established

Before this corpus was assembled, a focused audit on a single complete year (2015) tested the design across the three databases. It is not re-run here – it is preserved, runnable, in `Scripts/2015-audit/` – but it settled four things that motivated the collection approach, all of which the full corpus below then confirms at scale:

1. **Coverage is near-total.** OpenAlex contained essentially all (98.9–100%) of the trust research the curated Scopus and Web of Science searches returned, so OpenAlex can serve as the discovery substrate and citation backbone.
2. **No single query retrieves it all.** The OpenAlex topic-union retrieved under a tenth of the curated trust OpenAlex itself contains, and Web of Science was largely a subset of Scopus – because trust is studied *inside* hundreds of substantive topics, not in one.
3. **The databases are complementary.** Each channel held in-scope works the others missed, so recall has to come from merging all three by DOI, with precision imposed afterwards by the classifier and the screen.
4. **The only-OpenAlex works are mostly recoverable.** About two-thirds of the works only OpenAlex returned existed in a curated database after all – missed by the keyword *query*, not absent from the *database* – which is what later makes the curated by-DOI backfill (section 1.5) worthwhile.

This notebook applies the same three searches (Notebook 2., section 1.4) to the whole period and reports the result not as a test but as the **assembled corpus** the rest of the project works from. To resolve publication-year boundary mismatches across databases, the pull spans 1999–2026; 1999 and 2026 are buffer years, and the analysis window is **2000–2025** (a work enters the window if *any* source dates it inside it). Of 75,302 unique works in the full pull, **71,108 fall in the analysis window**; the rest are dated only to a buffer year in every source.

Two terms recur throughout, with the same meaning as in the databases notebook (section 1.5). The **OpenAlex pull** is the set of works the twelve-topic search returned; the **OpenAlex database** is what OpenAlex holds regardless of any query. The same split applies to the curated side – the **curated searches** (the Scopus and Web of Science keyword queries) versus the **curated databases** (those two indexes in full). ‘Recoverable’ therefore means present in a curated database but not returned by the curated searches, and a work can sit in the OpenAlex-pull-only region yet still exist in a curated database – a distinction that matters directly for the abstract gap (section 3.10).

3.2 The collection

Each search is deduplicated within its own source before any comparison.

Table 3.1: Works returned by each search, 2000–2025 analysis window (deduplicated within each source).

Database	Records
Scopus	48,626
Web of Science	35,444
OpenAlex pull (topic-union)	23,807

Scopus is the broadest keyword search, Web of Science the more selective, and the OpenAlex twelve-topic union – which captures the trust-*defining* core rather than trust-as-studied-everywhere – the smallest.

3.3 Coverage: does the OpenAlex database contain the curated records?

For every Scopus and Web of Science record with a DOI, the by-DOI crosswalk asks whether that DOI exists anywhere in the OpenAlex database – a query-independent check, read offline from the topic pull plus the crosswalk.

Table 3.2: Presence of each curated record in the OpenAlex database, full corpus.

Source	DOI'd records	In nAlex	Ope- nAlex %	Genuinely absent	Absent %	No-DOI
Scopus	45,898	45,179	98.4	719	1.6	2,728
Web of Science	34,326	34,007	99.1	319	0.9	1,118

The single-year result holds across the quarter-century: genuine absence is **1.6%** of DOI'd Scopus records and **0.9%** of Web of Science (an upper bound – a few malformed DOIs fail to resolve). **The OpenAlex database contains essentially all the trust research the curated searches return**, so it can serve as the discovery substrate and citation backbone, with curated records recoverable by DOI.

3.4 Overlap and uniqueness

Coverage is near-total, but the *retrieved* sets overlap modestly.

Table 3.3: Pairwise retrieval overlap across the window corpus.

Comparison	Overlap
Scopus found in OpenAlex	5,480 (11.3%)
WoS found in OpenAlex	4,687 (13.2%)
WoS found in Scopus	30,843 (87%)
Scopus found in WoS	30,843 (63.4%)

The OpenAlex pull retrieves about 11–13% of the curated trust the database overwhelmingly contains, and Web of Science is largely a subset of Scopus (87%) – both curated searches run the same precise query, while the OpenAlex pull draws a different boundary. *Why* each search returns less than its database holds was settled at the single year (section 3.1) and is consistent with the corpus: the dominant filter is substantive – trust is scattered across topics outside the twelve-topic net – with document-type exclusions accounting for much of the reverse gap.

3.5 The Venn regions: does OpenAlex’s classifier add value?

Each work sits in one of seven Venn regions and is triaged by the trust-sense classifier (Notebook 2.) into in-scope, out-of-scope or undetermined.

Table 3.4: The window corpus by Venn region and trust_scope.

region	n	in-scope	out-of-scope	undetermined
only OpenAlex	17,881	4,919	197	12,765
only Scopus	16,544	8,756	1,048	6,740
only WoS	4,155	2,574	126	1,455
OpenAlex+Scopus	1,239	915	20	304
OpenAlex+WoS	446	335	4	107
Scopus+WoS	26,602	16,785	734	9,083
all three	4,241	3,499	31	711
Total	71,108	37,783	2,160	31,165

The added-value finding replicates with force. OpenAlex’s topic search found **17,881 trust works that both curated keyword searches missed**; just 1.1% are out-of-scope, 27% are in-scope, and the rest await the content screen. The conclusion is bidirectional – the only-Scopus region holds 8,756 in-scope works the other two searches miss, the only-WoS region 2,574 – so no single channel is sufficient; the curated keyword searches contribute the largest blocks of uniquely-relevant recall, and works found in all three are the most clearly in-scope (83%).

3.6 The only-OpenAlex region: recoverable or genuinely unique?

The Venn table shows OpenAlex’s topic search surfaced 17,881 works the curated searches missed, but not whether those works are *genuinely* outside the subscription databases or merely missed by the curated *queries*. A reverse check settles it, exactly as at the single year (section 3.1): for every only-OpenAlex work with a DOI, the live Scopus and Web of Science APIs were asked whether the DOI exists in their database at all ([Scripts/collect/reverse-check.py](#)), capturing the curated document type where it does.

Table 3.5: The only-OpenAlex region split by the reverse check against the curated databases.

Of the only-OpenAlex works	n
with a DOI, in Scopus	9,459
with a DOI, in Web of Science	7,239
with a DOI, in either curated database (recoverable)	9,946
with a DOI, in neither (genuinely OpenAlex-unique)	5,672
no DOI (unverifiable)	2,263

Table 3.6: Curated document type of the recoverable only-OpenAlex works (top genres).

curated document type	n
Article	7,288
Book Chapter	616
Conference Paper	505
Review	366
Editorial material	333
Meeting	290
Abstract	176
Article; Meeting	94

The single-year pattern holds at scale. Of the only-OpenAlex works with a DOI, **9,946 (64%) are recoverable** – they exist in Scopus or Web of Science, the curated *queries* simply did not return them – and only **5,672 (36%) are genuinely outside both subscription databases**, OpenAlex’s true unique contribution. The recoverable works’ curated document types are dominated by **Article (7,288) and Book Chapter (616)** – exactly the types the curated query accepts, so these are genuine in-scope articles the keyword or field match failed to return – with the genres the query excludes by document type (Conference Paper 505, Review 366, Editorial 333, Meeting 290) a substantial but minority share that OpenAlex admits under its coarse “article” type. These 9,946 recoverable works are precisely what the curated by-DOI backfill (section 1.5) then enriches with curated abstracts and keywords; what that recovers is reported in section 3.10.

3.7 Where the trust signal sits in each database

A question that bridges to whatever screens the corpus next: in which field of each database’s record does the trust term actually appear? The classifier and any abstract-level screen can read a match in the title or abstract directly; a match only in the keyword fields – Web of Science’s cited-reference-derived Keywords Plus above all (Notebook 1.) – is findable but needs the keyword fields read; and a work a search returned on a trust term now absent from its held title, abstract and keywords is the abstract-withdrawal tail made visible. The table scans each returned record’s held fields for the trust word family (computed inline, not stored in the dataset; it approximates the finer categories of the single-year audit).

Table 3.7: Where the trust term sits in each source’s held record (works that source returned).

Source	returned	title / abstract	keywords only	elsewhere / withheld
OpenAlex pull	23,807	21,432	184	2,191
Scopus	48,626	46,330	578	1,718
Web of Science	35,444	33,428	1,018	998

For most records the trust signal is in the title or abstract, where any screen can read it. The exceptions are instructive. **Web of Science carries the most keyword-only matches (1,018)** – its Keywords Plus, generated from the titles of a work’s cited references, retrieves studies that engage the trust literature heavily without naming trust in their own text. **OpenAlex shows the largest “elsewhere/withheld” tail (2,191)**: works its topic search returned on a trust term that is no longer present in the held title, abstract or keywords – the abstract drought (Notebook 1.) made visible, since the search index remembered a match the record no longer exposes. Both are arguments for retaining the curated keyword fields and for the abstract backfill: the trust signal is not always where a naive title-and-abstract screen would look.

3.8 How trust research has grown, 2000–2025

The corpus shows what a single year cannot: the trajectory of the field and of each search’s grip on it.

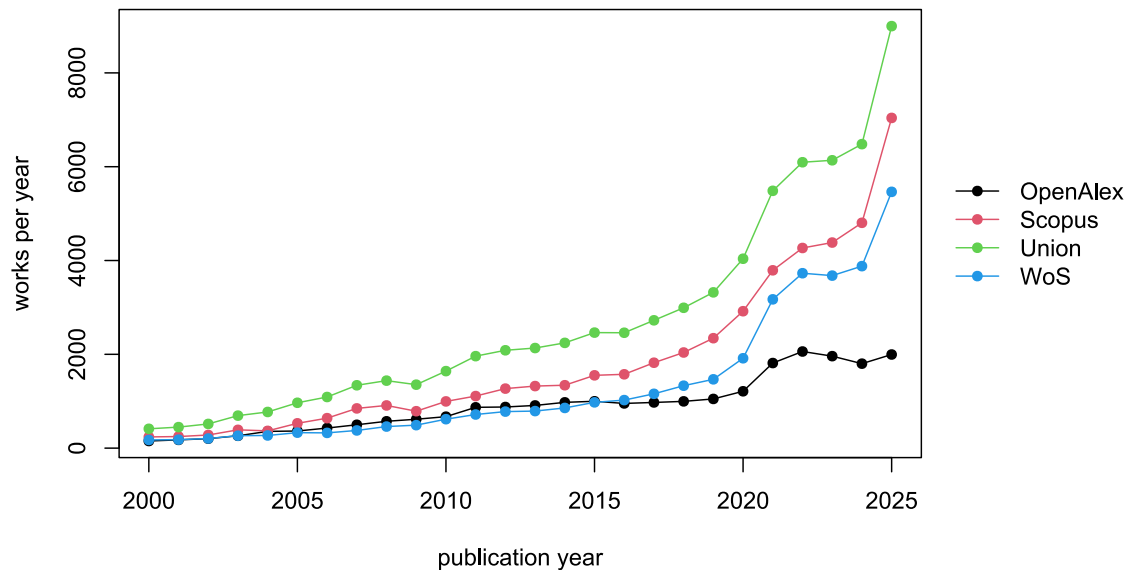


Figure 3.1: Trust-research works per year, 2000–2025: the deduplicated union and the share retrieved by each database search. The 2025 figure is inflated by online-first/in-press records and is read as a lower bound on the most recent year.

The same years, split by the trust-sense triage, show how the *relevant* share has held up as the field grew.

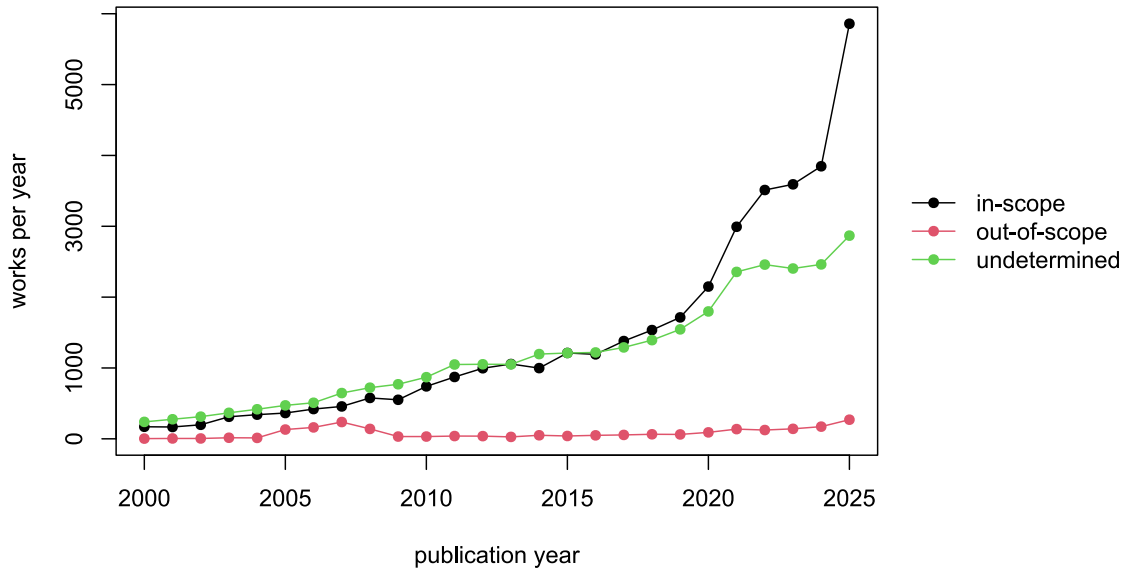


Figure 3.2: Trust-research works per year by trust_scope, 2000–2025: in-scope, undetermined and out-of-scope. The in-scope share holds steady across the period, so the growth is growth in relevant trust research, not in noise.

Three things stand out. The field has grown roughly twentyfold – from about 412 works in 2000 to several thousand a year by the early 2020s. The curated searches scale with that growth while the OpenAlex topic-union flattens (it levels off around 1,000–2,000 works a year and dips in the most recent years), consistent with a fixed twelve-topic net capturing a shrinking fraction of a widening field and with the abstract drought eroding the keyword clause OpenAlex depends on (Notebook 1.). The 2025 column is depressed for OpenAlex and inflated for the curated databases by online-first effects at the collection boundary, so the final year is read as provisional. And throughout, the **in-scope share holds steady** (Figure 3.2): the growth is growth in relevant trust research, not in noise, with out-of-scope work a thin, roughly constant band.

3.9 The trust-sense triage across the corpus

The record-level classifier (Notebook 2.) labels every work’s `trust_scope` and records where the sense came from. Across the window corpus:

Table 3.8: The trust_scope triage over the window corpus, and where each resolved sense came from

trust_scope	n	%
in-scope	37,783	53.1
out-of-scope	2,160	3.0
undetermined	31,165	43.8
Total	71,108	99.9

Table 3.9: Where each work’s combined sense came from (none = undetermined)

sense_basis	n
both	8,779
keywords	2,135
none	31,165
text	29,029
Total	71,108

The rule places **53.1%** of window works in-scope, **3.0%** out-of-scope, and leaves **43.8%** undetermined for the screen. Reading the keyword fields is not cosmetic: **2,135 works were classified from their keywords alone** (their title and abstract said nothing the rule could catch), and a further 8,779 had text and keywords agree; only **213 works** carry a scope conflict – a small, explicit review queue rather than a silent guess. The leading in-scope senses are political, institutional and generalised/social trust; the out-of-scope side is dominated by technology and consumer trust, exactly the contamination the triage is meant to hold back. Undetermined is the large residual by design (Notebook 2.) – it is the screen’s main queue, not a verdict of irrelevance.

3.10 What is still without an abstract

The classifier and any downstream screen depend on the abstract, so the works that lack one are worth characterising. The curated by-DOI backfill (section 1.5) has already run: of the 9,946 recoverable only-OpenAlex works it recovered curated metadata for, it closed most of the abstract gap, and the residual no-abstract set is now **1,859** works (down from 3,278 before the backfill).

Table 3.10: Characteristics of the works still with no abstract from any source (analysis window).

characteristic	value
document type (top 2)	article 1,143; book chapter 712
only in the OpenAlex pull (not returned by a curated search)	1,802
OpenAlex venue type (top 3)	journal 757; (none) 707; ebook platform 216

The residual is **97% (1,802) pull-only** works. Splitting them by the reverse check (Table 3.5) shows what the backfill left behind:

Table 3.11: The residual abstract-less pull-only works split by the reverse check (after the curated backfill).

Of the residual abstract-less pull-only works	n
with a DOI, in a curated database (no abstract there either)	205
with a DOI, in neither (genuinely OpenAlex-unique)	1,098
no DOI (unverifiable)	499
Total	1,802

Before the backfill, about 1,629 of the abstract-less pull-only works were recoverable; the curated pull supplied an abstract for most of them, leaving just **205** whose curated record carried no abstract either (conference and meeting items, mostly). What remains is overwhelmingly the **genuinely OpenAlex-unique tail** (1,098, in neither curated database) and the **no-DOI tail** (499) – works that cluster in book series, sourceless venues and ebook platforms rather than in the reputable journals, exactly where an

abstract is least likely to exist anywhere. All are retained and flagged, not dropped; an open-access full-text channel at the META stage is the remaining route to the few that have one.

3.11 The assembled dataset

The pipeline writes the analysis dataset to `data/final/corpus_derived.parquet`; the assembled base before classification is `corpus_raw.parquet` (same folder), the intermediate spine and the OpenAlex citation graph live in `data/intermediate/`, and every column is documented in `codebook.html` (Notebook 5., generated from the derivation registry). Parquet is the canonical format (read directly by these notebooks via R `arrow`); a CSV can be produced from any dataset on demand in one line of R or Python (see the RA guide). Of the 71,108 window works, **69,249 (97.4%) carry an abstract** – up from 95.4% before the curated backfill – of which **52,353 have an openly-licensed copy available** (`abstract_open_available`, i.e. an OpenAlex abstract exists) and the rest are licensed-only; each abstract is tagged by source, licence and whether it was backfilled, so external-API eligibility and any future redistribution are decided per work rather than by discarding data. The citation graph (2.77M edges) supports the reference miner; it is kept out of the working dataset so the corpus stays lean.

3.12 What the corpus establishes

1. **Coverage holds at scale.** The OpenAlex database contains 98.4–99.1% of what the curated searches return across 2000–2025; it serves as the discovery substrate and citation backbone.
2. **Discovery is multi-channel by necessity.** The topic-union retrieves about an eighth of the curated trust it contains, and that fraction shrinks over time; recall comes from complementary channels merged by DOI.
3. **The classifier’s value is real and large.** The OpenAlex search surfaces 17,881 in-scope-dense, low-noise trust works the keyword searches miss; the curated searches contribute the largest blocks of uniquely-relevant recall. Precision is imposed downstream by the trust-sense triage, the document-type detail, and the content screen.
4. **The curated backfill closed most of the abstract gap.** Recovering curated records for the 9,946 recoverable pull-only works lifted abstract coverage to 97.4% and enriched the corpus with curated keywords and document types, leaving a residual of 1,859 works that genuinely lack an abstract anywhere.

The 2015 audit chose the design; the 2000–2025 corpus confirms it and is the dataset the screening, extraction and reduction stages now operate on. Every column is catalogued in the codebook (Notebook 5.).

4. The THEME pipeline

i Status: visual summary

This notebook is a single picture of how the THEME corpus was built, from the raw searches to the analysis dataset and its codebook. It summarises what the preceding chapters describe in prose; every stage shown has run and produced the counts quoted (June 2026 build). The downstream stages – per-paper extraction, the reduction/review samples, the reference-based “gem” miner, and the versioned release – are deliberately left out here so the picture stays a clean summary of the dataset that exists; they are taken up separately once they are built.

4.1 The pipeline at a glance

Three searches are collected and preserved raw, assembled into one deduplicated corpus (with the curated DOI-backfill folded in as metadata), then enriched with the analytical variables. The design itself was settled by the 2015 audit (section 3.1) and confirmed on the full corpus (Notebook 3.).

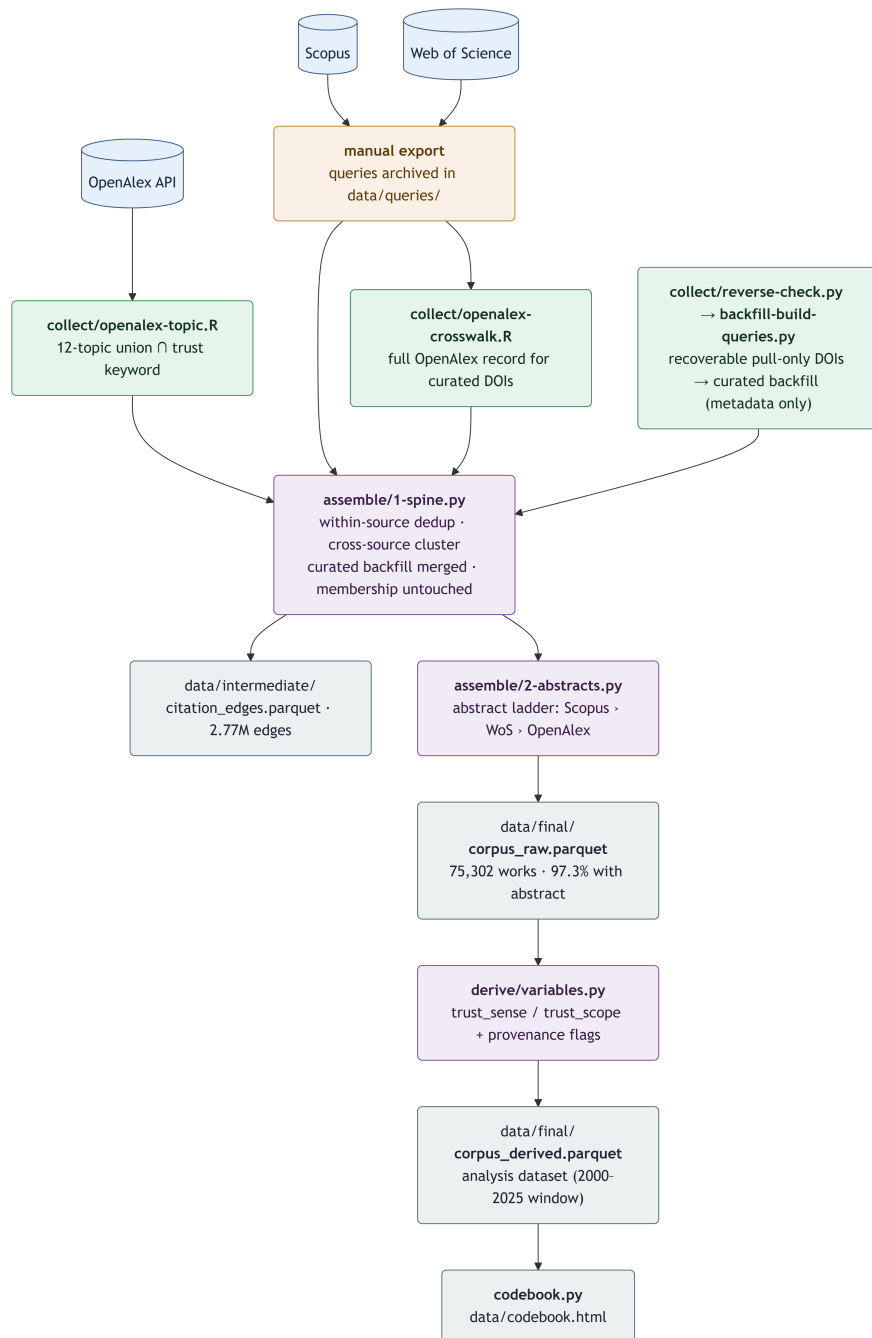


Figure 4.1: The THEME construction pipeline: collection (raw, 1999–2026) → assembly → derivation → the analysis dataset and codebook. Colour marks the kind of step: blue a source, green an automated script, amber a manual export, purple an assembly/derivation step, grey an output.

4.2 Reading the pipeline

💡 Collection (Stage A) – preserved raw, 1999–2026

Three searches run in parallel, each kept exactly as exported so the raw layer is reproducible. The **OpenAlex topic pull** (`collect/openalex-topic.R`) retrieves the union of twelve curated trust topics intersected with the trust keyword, by API. **Scopus** and **Web of Science** are exported by hand (their interfaces cap programmatic abstract retrieval and their abstracts are licensed for local processing only); the exact queries are archived in `data/queries/` and set out in section 1.4. The **OpenAlex crosswalk** (`collect/openalex-crosswalk.R`) fetches the full OpenAlex record for every curated DOI not already in the topic pull, so the citation graph attaches to the curated works too. Finally, the **curated DOI-backfill** (`collect/reverse-check.py` identifies the recoverable pull-only works; `collect/backfill-build-queries.py` builds the by-DOI queries) recovers curated abstracts, keywords and document types for the only-OpenAlex works that exist in a curated database (section 1.5). 1999 and 2026 are buffer years so 2000/2025 boundary mismatches resolve at merge.

💡 Assembly (Stage B) – one deduplicated corpus

`assemble/1-spine.py` deduplicates each source within itself, clusters the sources into one row per unique work (DOI, then exact normalised title), records which *searches* returned each work (`in_OpenAlex` / `in_WoS` / `in_Scopus`), and attaches the raw metadata from every source. The curated backfill is folded in here as **metadata only** – it fills the `Scopus_*` / `WoS_*` fields and sets the `*_backfilled` flags, but never changes the search-membership flags, so the coverage and added-value results stay intact. The OpenAlex citation graph is split out to `citation_edges.parquet` (kept out of the working corpus). `assemble/2-abstracts.py` then resolves the **abstract ladder** (Scopus > WoS > OpenAlex), tags each abstract's source, licence and whether it was backfilled, and writes `corpus_raw.parquet`.

💡 Derivation (Stage C) – the analysis dataset

`derive/variables.py` reads `corpus_raw`, applies the trust-sense classifier (`trust_context.py`, Notebook 2.) to compute `trust_sense` / `trust_scope` with their provenance and conflict flags, flags the 2000–2025 analysis window, and writes `corpus_derived.parquet` – the dataset every table in Notebook 3. is computed from. `codebook.py` renders the derivation registry (`derivations.yaml`) into `data/codebook.html`, documenting every column (Notebook 5.). Because derivation does no gathering, it is cheap to re-run whenever the classifier or the corpus changes.

The canonical format is **Parquet** (read directly by the notebooks via R `arrow`); a CSV can be produced from any dataset on demand in one line of R or Python (see the RA guide), so no CSV copies are kept in the repository. Abstracts are retained with source and licence flags, so external-API eligibility and any future redistribution are decided per work rather than by discarding data. *(The HTML edition of this diagram is intended to gain interactive, click-through annotations once the pipeline is final; the static figure above is the reference for both the HTML and PDF editions for now.)*

5. The THEME codebook

i Status: generated artefact

The codebook is generated from the derivation registry (`Scripts/derivations.yaml`) and the dataset itself by `Scripts/codebook.py`, which writes `data/codebook.html`. It documents **every** column of `corpus_derived` – its pipeline stage, the function that builds it, a description, fill rate and an example value – so provenance and documentation live in one place and cannot drift from the data.

`corpus_derived.parquet` is the analysis dataset (one row per unique work). Its columns fall into a small number of families: the **identity** and **membership** fields and the **derived** classification come from the assemble and derive stages and are documented one by one in the registry; the bulk of the width is the **raw source metadata** carried verbatim from each source (`OA_*`, `Scopus_*`, `WoS_*`), auto-documented by the codebook generator. The table below summarises the families; the full column-by-column codebook follows.

Table 5.1: Column families in `corpus_derived`.

Family	Columns
Web of Science (raw source)	72
Scopus (raw source)	33
OpenAlex (raw source)	28
Identity / year / document type / content	15
Membership + backfill provenance	14
Derived: trust classification	7
Abstract (resolved + flags)	4

The complete column-by-column codebook – every column with its stage, description, fill rate and an example value – is the generated HTML file `data/codebook.html`, best read in a browser; it is not reproduced inline in the PDF because of its length.

II

META – the meta-analytical dataset

III

METHOD – the methodological dataset

IV

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